

IN VITRO EFFICACY STUDY
**EVALUATION OF THE PROTECTIVE CAPABILITY FROM
BLUE LIGHT OF A COSMETIC PRODUCT ON CELL CULTURE**

**S & J INTERNATIONAL ENTERPRISES PUBLIC CO.,
LTD.**

KJELDGAARD BARRIER DEFENSE

SBW006-05

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REPORT CHANGE RECORD

The table here below reports the change log of all approved changes made to the document that make up the course after initial approval.

Rev. no	Date	Description
00	23/12/2022	Draft report release

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1. STUDY DESIGN

1.1 Title

In vitro efficacy study – evaluation of the antioxidant capability and of the protective capability from blue light of a cosmetic product on cell cultures.

1.2 Aim of the study

The study described in this report concerns with the in vitro evaluation of the capability of the tested product to protect keratinocyte cell cultures against the stress induced by blue light.

This evaluation was carried out by determination of cell viability and oxidative stress level after cell treatment with product under study and UV exposure, typical oxidizing agent.

Experimental protocol provided the following experimental conditions:

- untreated keratinocyte cell cultures (CTR-);
- keratinocyte cell cultures not treated and exposed to blue light diffused radiation by a LED light system (CTR+);
- keratinocyte cell cultures treated with tested product at three different concentrations (0,1%, 0,05% and 0,025%) and exposed to blue light radiation.

The evaluation of the efficacy of the test substance was achieved after 2 repeated day/night cycles exposure (exposures of 8 hours).

1.3 Tested product

S & J INTERNATIONAL ENTERPRISES PUBLIC CO., LTD.

**KJELDGAARD BARRIER DEFENSE
SBW006-05**

Tested item is a cosmetic product with the following INCI declared by the Sponsor:

Aqua (Water), Glycereth-26, Butylene Glycol, Bis-PEG-18 Methyl Ether Dimethyl Silane, Methyl Gluceth-20, PEG-8, Propanediol, Glycerin, Propylene Glycol, Ethylhexylglycerin, Phenoxyethanol, Bakuchiol, Ferulic Acid, Tocopheryl Acetate, Tromethamine, Carbomer, Bisabolol, Pentylene Glycol, Buddleja Officinalis Flower Extract, Trisodium Ethylenediamine Disuccinate, Squalane, Xanthan Gum, Lecithin, Hydrolyzed Collagen, Hydroxypropyl Cyclodextrin, Alcohol, Polysorbate 20, Retinol, Caprylyl Glycol, Sodium Acetylated Hyaluronate, Sodium Hyaluronate, Sodium Hyaluronate Crosspolymer, Potassium Phosphate, Hydrolyzed Sodium Hyaluronate, Acetyl Hexapeptide-8, BHT, Palmitoyl Tripeptide-38, BHA, Tocopherol, Ascorbyl Palmitate

1.4 Dates

Experimental session: 29/11/2022 - 02/12/2022

Biochemical dosages: 06/12/2022

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2. MATERIALS AND METHODS

2.1 Experimental model

The biological model used in the test consists of a human keratinocyte cell culture.

2.2 Preparation of the test item and cell exposure

For the test execution, keratinocyte cell culture was first treated with product under study and then exposed to blue light radiation. The keratinocyte cell culture was first treated with the test item at three different concentrations chosen after a preliminary cytotoxicity test (0,1%, 0,05% and 0,025%) and then exposed to 2 cycle day/night of blue light radiation. At the end of the exposures, the cells were washed with PBS and MTT assay was performed in order to evaluate the cell viability.

Furthermore, the cells were homogenate to perform the MDA dosage to quantify the lipoperoxidation damage.

The results were compared to negative control (untreated cells, CTR-) and positive control (CTR+).

In summary, the experimental protocol provided:

- untreated keratinocyte cell cultures (CTR-);
- keratinocyte cell cultures not treated and exposed to blue light diffused radiation by a LED light system (CTR+);
- cells treated with product KJELDGAARD BARRIER DEFENSE SBW006-05 and exposed to 2 day/night cycles with 8 hours of blue light exposure each one.

At the end of the exposures, the cells were washed with PBS and MTT assay was performed in order to evaluate the cell viability.

2.3 Cell viability study – MTT assay

The MTT (3,(4,5-dimethylthiazol-2),2,5 difeniltetrazolium bromide) test is a standard, simple and accurate colorimetric method for cell viability assessment. The assay is based on the intracellular reduction of the yellow tetrazolium salts by the mitochondrial enzyme succinate dehydrogenase in blue/purple formazan crystals. The reaction may therefore take place only in metabolically active cells and the value of the optical density obtained by photometric reading can be correlated to the amount of viable cells.

At the end of each treatment the wells were rinsed with PBS (phosphate-buffered saline solution), stained with MTT solution 1 mg/mL and incubated for three hours at 36,5°C/5% CO₂. Then the wells were treated with isopropanol and incubated for two hours at room temperature. After isopropanol incubation, absorbance readings were performed at 570 nm by microplate reader (isopropanol was used as blank for reading). For each test condition the ratio of the average optical density of the treated cultures on the average optical density of negative controls determines the viability rate. The protection %, intended as cell viability % increase compared to the positive control condition, treated only with the damaging agent, is also calculated.

2.4 Lipid damage study – MDA dosage

Malondialdehyde content determination was used as oxidative stress index related to the lipid component. Malondialdehyde is in fact a specific biomarker of oxidative stress for lipids specifically associated to the state of lipid peroxidation. This phenomenon begins at the level of polyunsaturated fatty acids of the membrane phospholipids; free radicals react with phospholipids by oxidizing them and thus lead to the formation of unstable lipid hydroperoxides that decompose producing a number of secondary products such as aldehydes and ketones recognized as toxic or carcinogenic substances. After the beginning and subsequent propagation of lipid peroxidation in biological membranes the products may in turn trigger new radical reactions resulting in a harmful biological ripple

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effect. Malondialdehyde (MDA) and 4-hydroxy-2-nonenal (4-HNE) are the two main products of lipid peroxidation, and their concentration in biological systems is a good index of lipo-peroxide damage.

To determine the lipoperoxides levels the colorimetric method tested by Erdelmeier and collaborators (1998) was assayed: the test is based on the capability of a chromogen, N methyl 2 phenylindole (NMPI), to react with MDA at 45°C with acid pH to produce a stable blue chromophore that has an absorption pick at 586 nm.

The quantitative determination uses a calibration curve made-up of known and growing concentrations of standard MDA. The results are expressed as MDA concentration (μM) in 100 μL cell homogenate. The difference in MDA content between positive control/sample and negative control is calculated.

2.5 Statistical analysis

Obtained results were subjected to statistical analysis by means of Student test. Variations (vs CTR+) are considered statistically significant with $p^* < 0.05$. The % variation versus CTR - was calculated for each experimental condition and then the % protection was calculated as difference between % variation values of CTR+ and cells treated with tested sample.

2.6 Bibliographic references

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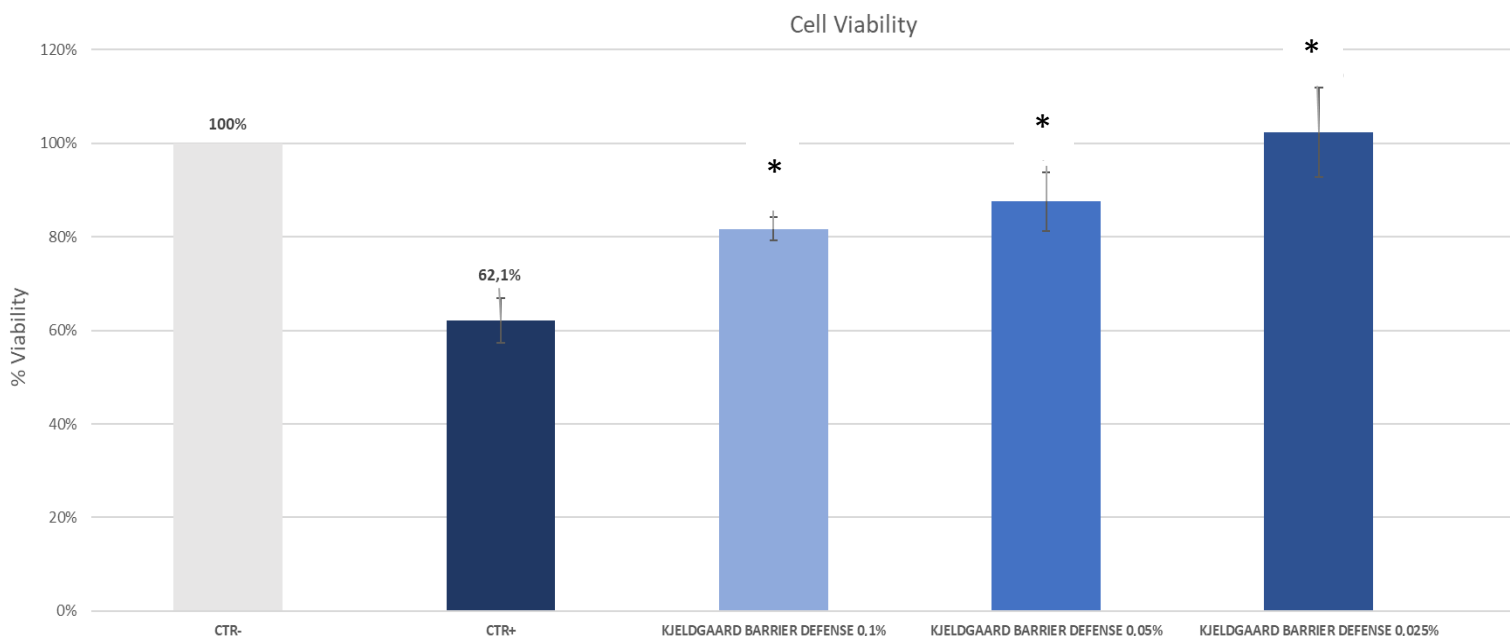
3 RISULTATI / RESULTS

The data obtained in this study are reported in the tables and graph below.

CELL VIABILITY

Results are expressed as mean viability $\pm$ st. dev. (expressed in % compared to CTR-) and % protection compared to CTR+.

Blue Light protection	Viability			
	Mean	St. dev.	Protection %	
CTR-	100%			
CTR+	62,1%	4,8%		
KJELDGAARD BARRIER DEFENSE 0,1%	81,7%	2,5%	19,5%	*
KJELDGAARD BARRIER DEFENSE 0,05%	87,5%	6,2%	25,4%	*
KJELDGAARD BARRIER DEFENSE 0,025%	102,4%	9,5%	40,2%	*



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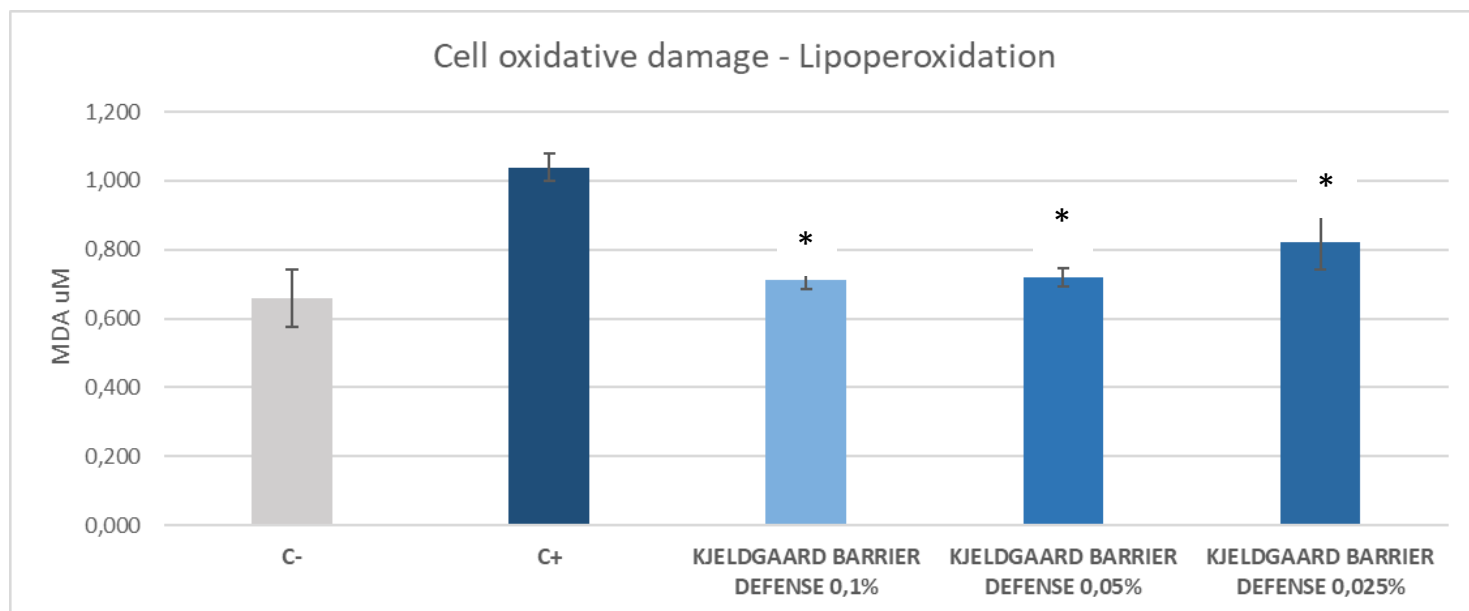
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LIPOPEROXIDATION

MDA dosage results expressed as mean $\pm$ st. dev. (expressed μ M MDA). % Variation of MDA content compared to CTR- and % protection compared to CTR+ were reported.

Blue Light protection	Cell oxidative damage - Lipoperoxidation				
	Mean	St. dev.	VAR% vs CTR-	Protection %	
CTR-	0,659	0,08363			
CTR+	1,039	0,039	57,7%		
KJELDGAARD BARRIER DEFENSE 0,1%	0,711	0,026	7,9%	49,7%	*
KJELDGAARD BARRIER DEFENSE 0,05%	0,719	0,027	9,2%	48,5%	*
KJELDGAARD BARRIER DEFENSE 0,025%	0,821	0,080	24,7%	33,0%	*



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Comment to the results

The exposure of cell culture to blue light irradiation showed a significant increase in oxidative damage and a decrease of cell viability ($p < 0.05$).

Keratinocytes cell cultures simultaneously treated KJELDGAARD BARRIER DEFENSE SBW006-05 and experimentally irradiated with blue light (2 day/night cycles with 8 hours of blue light exposure) showed a cell viability significantly higher compared to CTR+ and a significantly lower oxidative stress level ($*p < 0.05$) than the positive control one in all the three different concentrations (0,1%, 0,05% and 0,025%).

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4 CONCLUSIONS

According to the data obtained and reported in this report, we can assess that

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**KJELDGAARD BARRIER DEFENSE
SBW006-05**

***showed a significant protective capability against stress induced by blue light irradiation,
highlighting antioxidant efficacy and blue light protection***

Experimenter

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Study Director

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